

AI Product Intelligence System using CLIP and FAISS

1. Introduction

This project implements an AI-powered product intelligence system that performs reverse product search using multimodal AI. The system understands product images and text queries using CLIP embeddings and retrieves relevant products using FAISS vector search.

2. Dataset

The Fashion Product Images Small dataset from Kaggle was used. It contains fashion product images along with product metadata.

3. Technologies Used

- Python
- CLIP (Contrastive Language–Image Pre-training)
- FAISS Vector Database
- Hugging Face Transformers
- PyTorch

4. Methodology

1. Product images were processed using CLIP image encoder.
2. Each image was converted into a 512-dimensional embedding vector.
3. Text queries were converted into CLIP text embeddings.
4. FAISS was used to find the most similar products.
5. Top matching products were displayed.

5. System Architecture

User Text Query



CLIP Text Encoder



Text Embedding



FAISS Similarity Search



Image Embeddings Database



Top Matching Products

6. Results

The screenshot displays a Kaggle Notebook Editor interface. The main code cell contains the following Python code:

```
[17]: model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32")
processor = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")

print("✅ CLIP model loaded successfully!")
```

Below the code, there are several progress bars and status messages:

- Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
- config.json: 4.19k/? [00:00<00:00, 457kB/s]
- pytorch_model.bin: 100% 605M/605M [00:04<00:00, 201MB/s]
- model.safetensors: 100% 605M/605M [00:03<00:00, 390MB/s]
- Loading weights: 100% 398/398 [00:00<00:00, 663.20it/s, Materializing param=visual_projection.weight]
- CLIPModel LOAD REPORT from: openai/clip-vit-base-patch32
- Key | Status | |
- +-----+-----
- vision_model.embeddings.position_ids | UNEXPECTED | |
- text_model.embeddings.position_ids | UNEXPECTED | |
- Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- preprocessor_config.json: 100% 316/316 [00:00<00:00, 14.0kB/s]
- The image processor of type "CLIPImageProcessor" is now loaded as a fast processor by default, even if the model checkpoint was saved with a slow processor. This is a breaking change and may produce slightly different outputs. To continue using the slow processor, instantiate this class with "use_fast=False".

On the right side, there is a sidebar with "Add Input" and "6 Recently Viewed Datasets". The datasets listed are:

- Fashion Product Images (Sm...)
- data_labels_fashion_product...
- Indian Tourism Dataset
- Traveler Trip dataset

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File Edit View Run Settings Add-ons Help

Cancel Run Code Draft Session (2h:13m)

```
1.6052e-01, 2.6348e-01, 2.0365e-02, 4.0283e-01, -1.6824e-01,
-3.0848e-01, 4.3407e-01, -1.4309e-01, 1.7899e-01, -1.4208e-01,
1.0350e-01, 2.8794e-02, -4.0977e-02, -6.4386e-02, 2.4566e-01,
-4.9512e-01, 4.4275e-01, 3.4548e-01, -4.1417e-01, -1.4715e-01,
2.8705e-01, 3.5809e-02, 2.1146e-02, 1.8418e-01, 3.4729e-01,
1.1476e-01, -6.8257e-02, 4.9052e-01, -2.1651e-01, 3.6063e-01,
7.4866e-03, 3.8905e-02, -3.2263e-01, 2.5383e-02, 1.1496e-01,
6.3008e-01, 8.3095e-02, 1.6107e-01, -2.1479e-01, 2.5547e-01,
-2.8273e-01, 6.5918e-02, 2.3266e-01, 1.3322e-01, 4.4288e-01,
1.4446e-01, -8.2712e-01, -1.8829e-01, -2.6477e-01, 1.5094e-01,
3.5024e-01, -5.9020e-02, -5.6506e-03, 1.8639e-01, -2.7569e-01,
-4.5255e-02, -5.3516e-02, -1.3369e-01, 9.1783e-02, -2.3624e-01,
-2.1015e-02, 5.0204e-01, -4.7313e-01, 2.6237e-01, -1.0873e-01,
-1.4527e-01, 1.5821e-01, -5.5391e-02, -3.6669e-01, 6.0228e-01,
-3.0025e-01, -2.6455e-01, 4.3983e-01, -2.8589e-01, 7.6725e-01,
9.0144e-02, 2.0230e-01]]], hidden_states=None, attentions=None)
```

```
[21]: with torch.no_grad():
      outputs = model.get_image_features(**inputs)
      image_embedding = outputs.pooler_output
      print(type(image_embedding))
      print(image_embedding.shape)

      <class 'torch.Tensor'>
      torch.Size([1, 512])
```

Add Input

Search keyword or URL

All Your Work Datasets Models

Competition Datasets Notebook

Utility Scripts

6 Recently Viewed Datasets

Fashion Product Images (Sm...
Param Aggarwal · Updated 7y ago
477 Upvotes · 44441 Files (other, CSV) · 59...

data_labels_fashion_product...
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Indian Tourism Dataset
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12 Upvotes · 3 Files (JSON, other) · 128 kB

Traveler Trip dataset
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Search

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```
if os.path.exists(image_path):
    image = Image.open(image_path).convert("RGB")

    # Convert image for CLIP
    inputs = processor(
        images=image,
        return_tensors="pt"
    )

    with torch.no_grad():
        output = model.get_image_features(**inputs)

    # Extract 512 dimensional embedding
    embedding = output.pooler_output.cpu().numpy()

    # Store embedding and path
    image_embeddings.append(embedding[0])
    image_paths.append(image_path)

print("Total embeddings created:", len(image_embeddings))
print("Total image paths stored:", len(image_paths))
```

```
100%|██████████| 44424/44424 [1:11:53<00:00, 10.30it/s]
Total embeddings created: 44419
Total image paths stored: 44419
```

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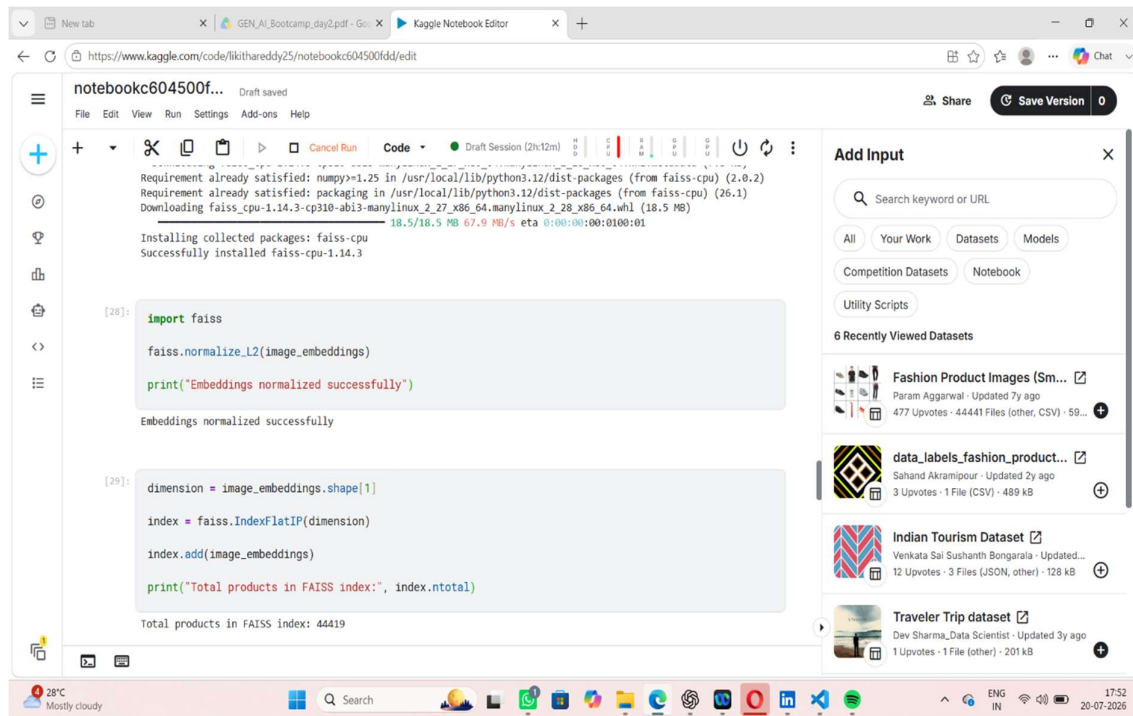
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The system was tested with different text queries and retrieved visually similar fashion products using CLIP embeddings and FAISS similarity search.

Query 1:

Write:

Query: "blue casual shirt"

Retrieved Products:

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```
print("Product ID:", product_id)
print("Product Name:", product_name)
print("-----")
```

HM Women Black Shoes



Product ID: 56992
Product Name: HM Women Black Shoes

Facit Men Maroon Comfort Briefs

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File Edit View Run Settings Add-ons Help

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Product ID: 56992
Product Name: HM Women Black Shoes

Facit Men Maroon Comfort Briefs



Product ID: 15904
Product Name: Facit Men Maroon Comfort Briefs

Arrow Woman Blue Shirt

Add Input

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Product ID: 15984
Product Name: Facit Men Maroon Comfort Briefs

Arrow Woman Blue Shirt



Product ID: 32393
Product Name: Arrow woman Blue shirt

Probase Black Printed Boxers



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File Edit View Run Settings Add-ons Help

Cancel Run Code Draft Session (2h:30m)

Product ID: 32393
Product Name: Arrow woman Blue shirt

Probase Black Printed Boxers



Product ID: 32357
Product Name: Probase Black Printed Boxers

Amante Peach-Coloured T-shirt Bra BFCV22



Add Input

Search keyword or URL

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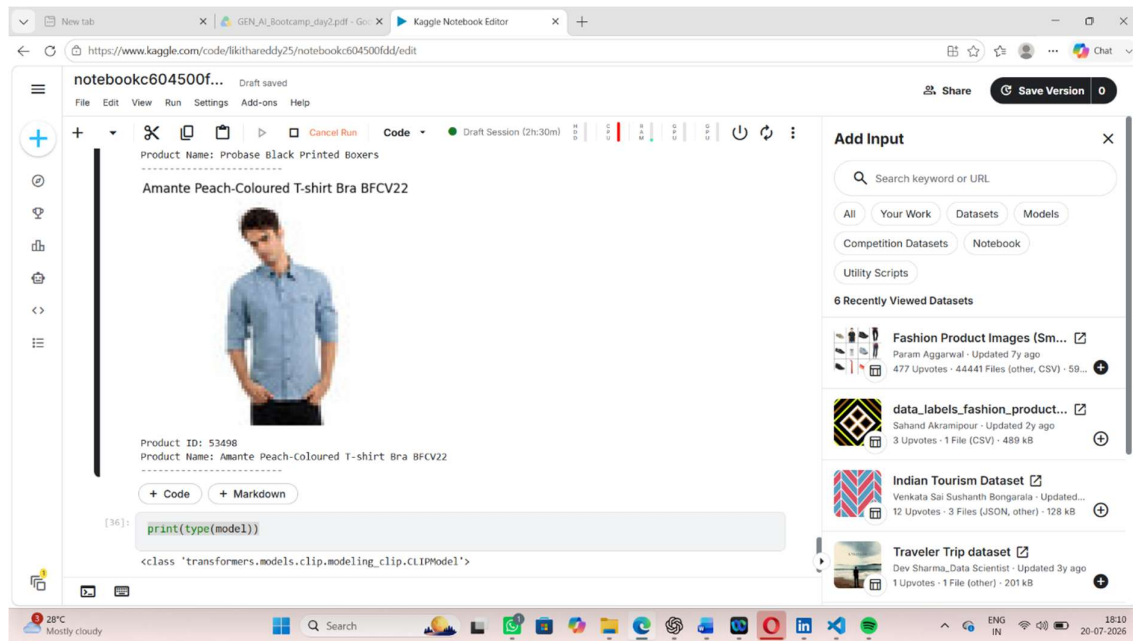
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7. Conclusion

The AI Product Intelligence System successfully performs reverse product search using multimodal AI techniques. CLIP embeddings were used to understand the relationship between text queries and product images, while FAISS enabled efficient similarity-based retrieval. The system can retrieve relevant fashion products based on natural language queries.

8. Future Improvements

Future improvements can include personalized recommendations, product price comparison, seller-based ranking, and integration with real-time e-commerce platforms.